

LLM RAG Sample Test Results

Project: eCOM-AI | Date: 2026-04-25 | Endpoint: /api/ai/chat

Summary: 14 sample questions were tested after adding the LLM answer-generation layer. Product and delivery questions returned natural customer-facing answers. Unsupported questions were rejected before LLM generation, keeping the assistant grounded in store data.

What Was Tested

The test verified the full RAG flow:

question -> retrieval -> intent filtering -> structured filtering/ranking -> source coverage check -> LLM answer generation

Accepted Questions

#	Question	Decision	Answer	Sources
1	Which delivery option is cheapest?	Accepted	The cheapest delivery option is FREE delivery, which costs \$0.00.	FREE delivery UPS3 delivery UPS2 delivery
2	Which delivery option is fastest?	Accepted	The fastest delivery option is UPS1 delivery, with an estimated delivery time of 1-2 days. It costs \$10.00.	UPS1 delivery UPS3 delivery UPS2 delivery
3	What delivery options are available?	Accepted	Available delivery options: - UPS1 - 1-2 days, \$10.00 - UPS2 - 2-5 days, \$5.99 - UPS3 - 5-10 days, \$2.55	UPS1 delivery UPS2 delivery UPS3 delivery
4	Do you have free delivery?	Accepted	Yes - free delivery is available. - FREE delivery: \$0.00 - Estimated delivery time: 1-2 weeks Other delivery options are also available if needed.	FREE delivery UPS3 delivery UPS2 delivery
5	Show me hats	Accepted	Here are the hats I found: - Green React Woolen Hat - \$8.00 - Purple React Woolen Hat - \$15.00 - Core Blue Hat - \$10.00	Green React Woolen Hat Purple React Woolen Hat Core Blue Hat
6	Which hat is cheapest?	Accepted	The cheapest hat is Green React Woolen Hat at \$8.00.	Green React Woolen Hat Core Blue Hat Purple React Woolen Hat

#	Question	Decision	Answer	Sources
7	What products are available under 20 dollars?	Accepted	Products under \$20 found in the store data: - Green Code Gloves - \$15.00 - Blue Code Gloves - \$18.00 - Green React Gloves - \$14.00	Green Code Gloves Blue Code Gloves Green React Gloves
8	Show me React products	Accepted	Here are the React products I could find: - React Board Super Whizzy Fast - \$250.00 - Green React Woolen Hat - \$8.00 - Green React Gloves - \$14.00	React Board Super Whizzy Fast Green React Woolen Hat Green React Gloves
9	Do you have blue hats?	Accepted	Yes - we have a Core Blue Hat for \$10.00.	Core Blue Hat Purple React Woolen Hat Green React Woolen Hat

Rejected / Unsupported Questions

#	Question	Decision	Answer	Sources
10	Do you offer cash on delivery?	Rejected	I found related information, but I could not find a source that clearly confirms this. Try asking about a product, delivery option, price, or brand that is listed in the store.	None
11	Can I get next day delivery?	Rejected	I found related information, but I could not find a source that clearly confirms this. Try asking about a product, delivery option, price, or brand that is listed in the store.	None
12	Do you have discounts on hats?	Rejected	I found related information, but I could not find a source that clearly confirms this. Try asking about a product, delivery option, price, or brand that is listed in the store.	None
13	Do you ship internationally?	Rejected	I found some related information, but not enough to answer confidently. Try asking in a more specific way.	None
14	What warranty comes with laptops?	Rejected	I found some related information, but not enough to answer confidently. Try asking in a more specific way.	None

Result

Pass: The LLM layer generated natural answers only after retrieval and validation succeeded. The rejected cases did not reach a fabricated answer, which confirms the grounding checks are still protecting the assistant.